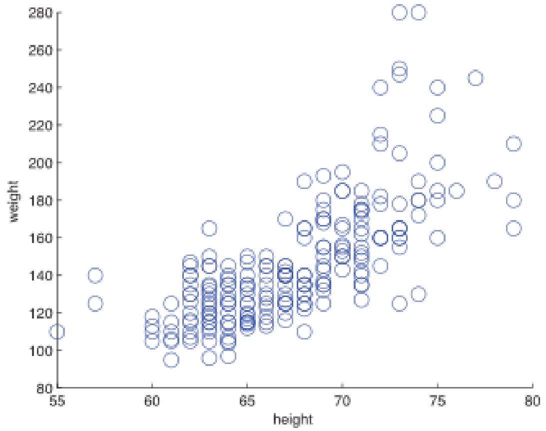
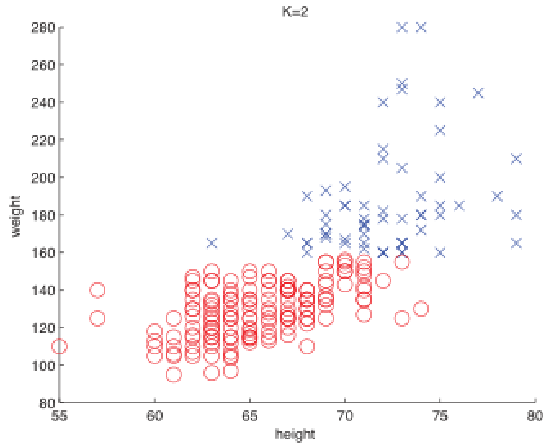
3. Mixture Models and the EM Algorithm

1. Mixture Models
2. EM Algorithm
3. Some Practice Problems
4. Background Reading: Theoretical Basis (Supplementary)

Motivation



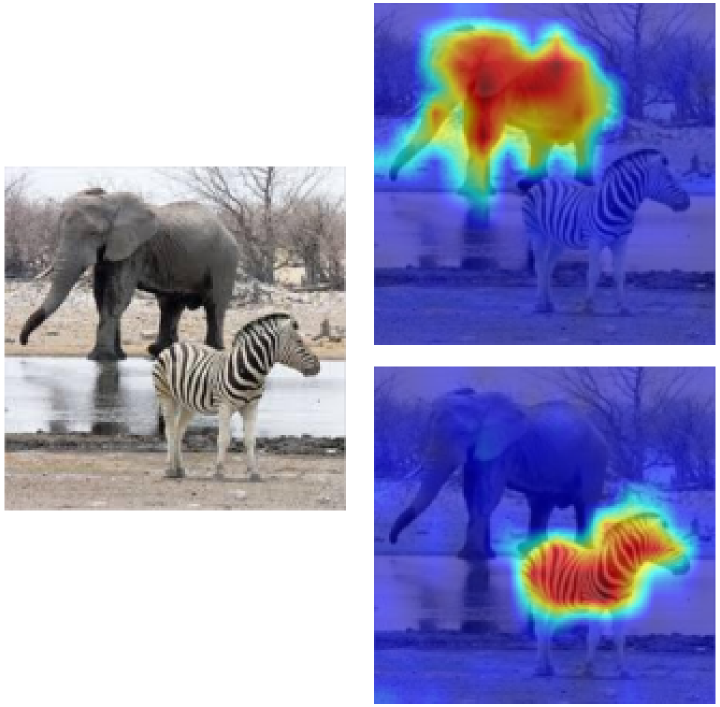
(a)

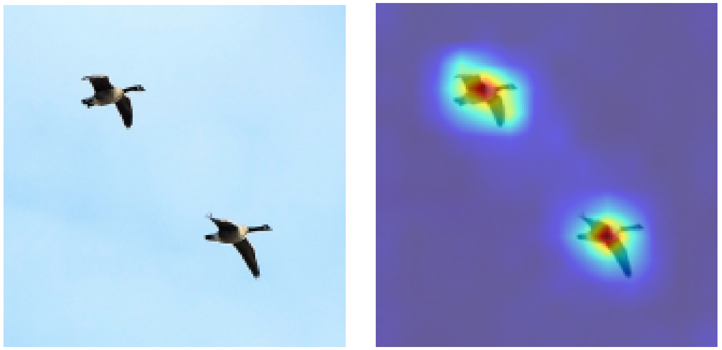


(b)

Height and weight of a population. How to model this bivariate distribution?

Motivation

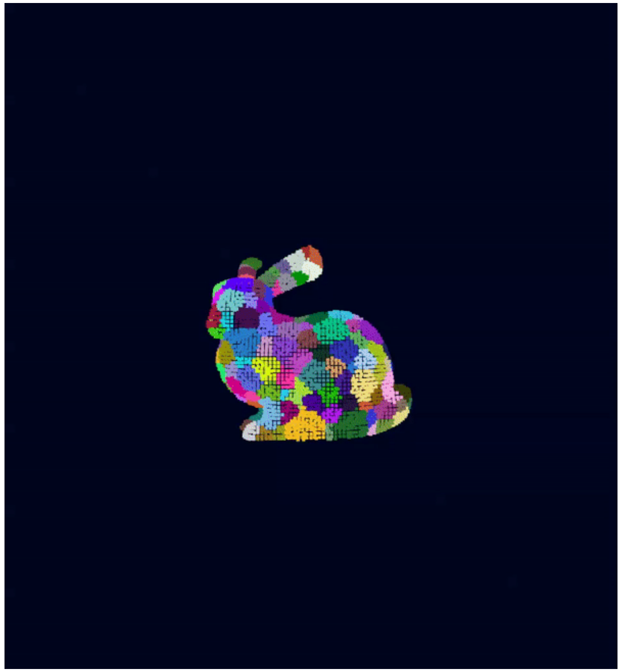


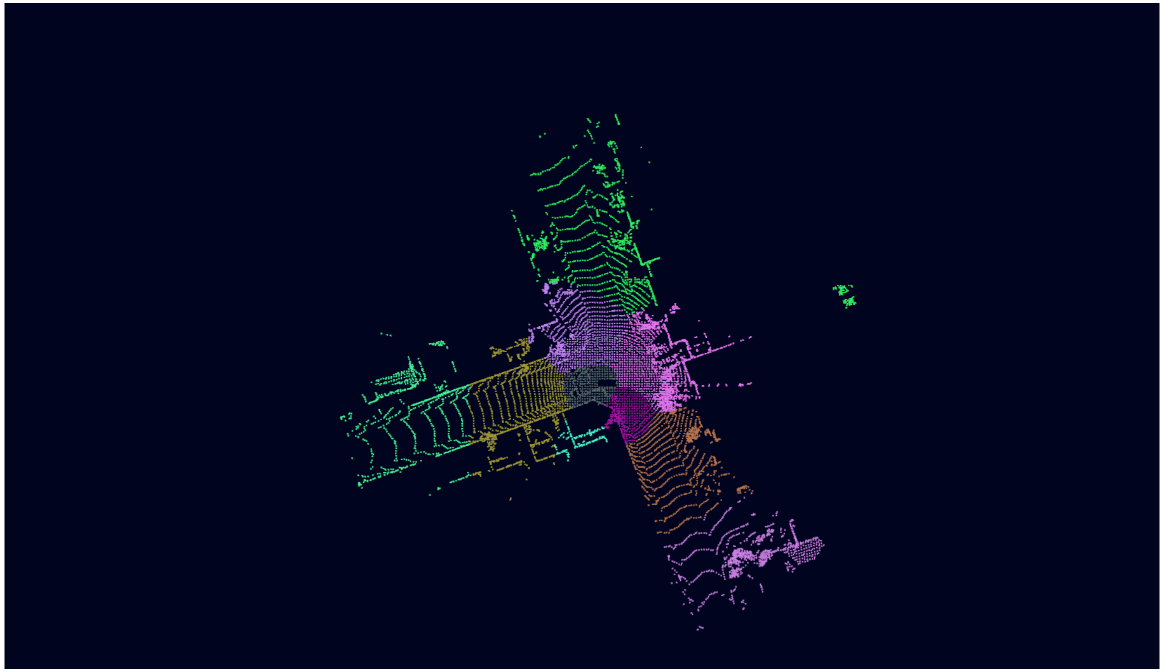


Motivation

https://github.com/somanshu25/GPU-Accelerated-Point-Cloud-Registration-Using-Hierarchical-GMM

Each color corresponds to a "mixture component". Bunny: 100 components, LiDAR: 50 components





Outline

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Latent Variable and Mixture

- Suppose an observation $\mathbf{x}$ can be generated from K possible pdfs $p(\mathbf{x} \mid \boldsymbol{\eta}_1), \ldots, p(\mathbf{x} \mid \boldsymbol{\eta}_K)$. Which one is unknown or unobserved.
- Let z be the index of the generating pdf. This can be modeled as a random variable with $p(z) = \mathrm{Cat}\,(z \mid \boldsymbol{\pi})$.
- As it is unobserved, we call it a *latent* variable. Then,

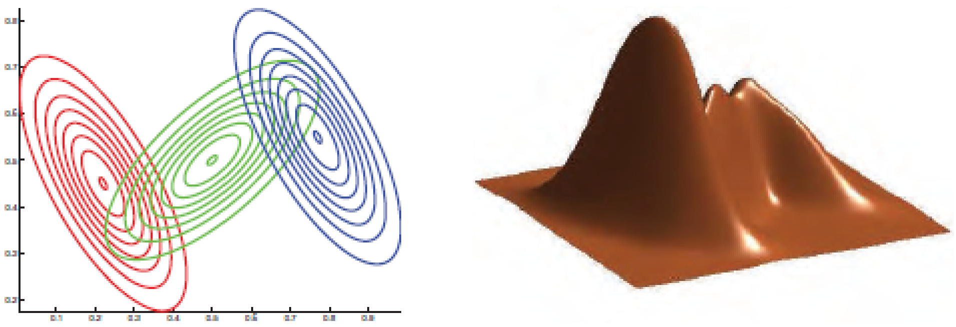
$$p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} p(\mathbf{x}, z = k \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k) p(\mathbf{x} \mid \boldsymbol{\eta}_k), \tag{3.1}$$

where $\boldsymbol{\theta} = (\boldsymbol{\pi}, \{\boldsymbol{\eta}_k : k = 1, \ldots, K\})$.

Example: GMM

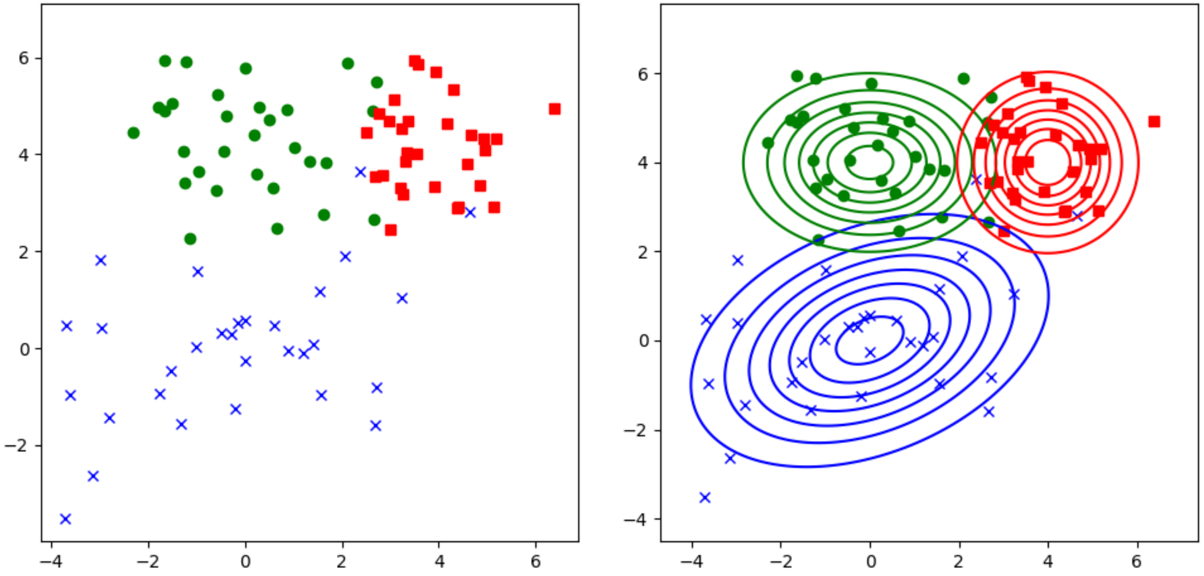
- Gaussian mixture model (GMM):

$$p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k)\mathcal{N}\left(\mathbf{x} \,|\, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\right), \qquad \boldsymbol{\theta} = (\boldsymbol{\pi}, \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^{K}). \tag{3.2}$$



(a) (b)

Example: GMM

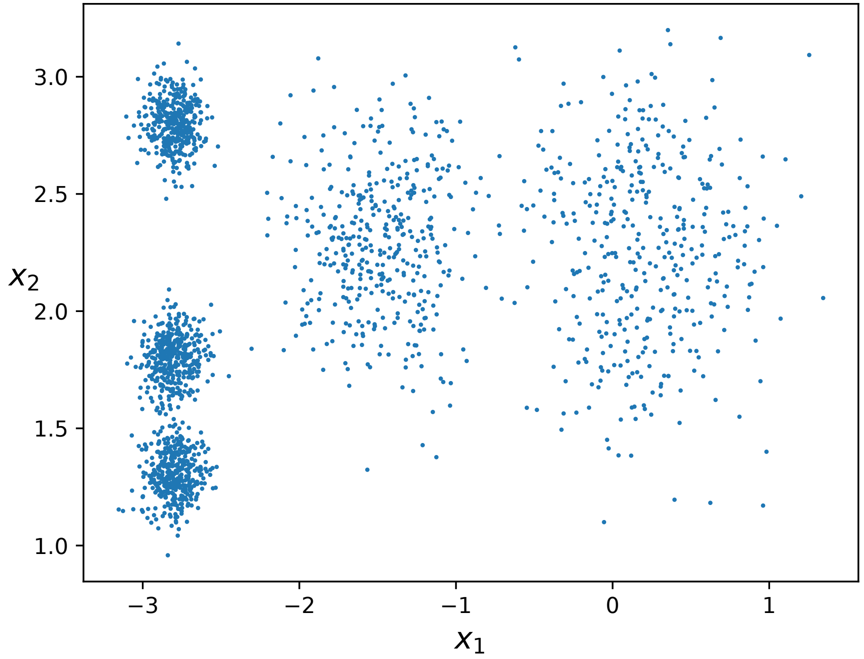


$$p(\mathbf{x} \mid \boldsymbol{\theta}) = 0.3\mathcal{N}\left(\mathbf{x} \,\middle|\, \boldsymbol{\mu}_1 = \left[\begin{smallmatrix}0\\0\end{smallmatrix}\right], \boldsymbol{\Sigma}_1 = \left[\begin{smallmatrix}4&1\\1&2\end{smallmatrix}\right]\right)$$
$$+ 0.3\mathcal{N}\left(\mathbf{x} \,\middle|\, \boldsymbol{\mu}_2 = \left[\begin{smallmatrix}0\\4\end{smallmatrix}\right], \boldsymbol{\Sigma}_2 = \left[\begin{smallmatrix}2&0\\0&1\end{smallmatrix}\right]\right)$$
$$+ 0.4\mathcal{N}\left(\mathbf{x} \,\middle|\, \boldsymbol{\mu}_3 = \left[\begin{smallmatrix}4\\4\end{smallmatrix}\right], \boldsymbol{\Sigma}_3 = \left[\begin{smallmatrix}2&0\\0&2\end{smallmatrix}\right]\right)$$

$$\boldsymbol{\pi} = \begin{bmatrix}\boldsymbol{\pi}(1)\\ \boldsymbol{\pi}(2)\\ \boldsymbol{\pi}(3)\end{bmatrix} = \begin{bmatrix}0.3\\0.3\\0.4\end{bmatrix}$$
$$\boldsymbol{\theta} = (\boldsymbol{\pi}, \{\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k\}_{k=1}^{3}).$$

Example: Clustering

- : 03_kmeans_voronoi.ipynb



- $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Model $p(\mathbf{x} \mid \boldsymbol{\theta})$ as a mixture of $K = 5$ Gaussians.

MLE

- Observations $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ are i.i.d.: $p(\mathbf{x}_i \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k) p(\mathbf{x}_i \mid \boldsymbol{\eta}_k)$
- The MLE $\boldsymbol{\theta}_{\mathrm{ML}}$ is found by maximizing the log likelihood

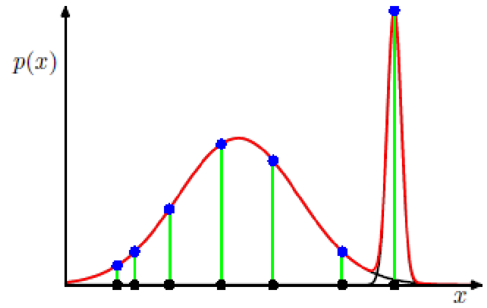
$$\log p(\mathbf{x}_1, \ldots, \mathbf{x}_n \mid \boldsymbol{\theta}) = \sum_{i=1}^{n} \log \sum_{k=1}^{K} \boldsymbol{\pi}(k) p(\mathbf{x}_i \mid \boldsymbol{\eta}_k).$$

Algorithmic Issues: Singularities

- Example, GMM: $p(\mathbf{x}_i \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k)\mathcal{N}\left(\mathbf{x}_i \,|\, \boldsymbol{\mu}_k, \sigma_k \mathbf{I}\right)$.
- Suppose $K > 1$. If for one of the k, choose $\boldsymbol{\mu}_k = \mathbf{x}_i$ and $\sigma_k \to 0$ ("collapsing"),

$$\mathcal{N}\left(\mathbf{x}_i \,|\, \boldsymbol{\mu}_k, \sigma_k \mathbf{I}\right) \propto \frac{1}{\sigma_k} \to \infty,$$

a singularity in the likelihood function!



- Heuristics to avoid collapse: reset a mean to random value, reset covariance

Algorithmic Issues: Unidentifiability

- We want models that are identifiable: for all $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$, $P_{\boldsymbol{\theta}_1} \neq P_{\boldsymbol{\theta}_2}$. To interpret the parameters found.
- MLE for mixture model: $p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k) p(\mathbf{x} \mid \boldsymbol{\eta}_k)$ — for any parameter $\boldsymbol{\theta}_{\mathrm{ML}}$, can permute the indices k to obtain another solution that gives the same overall pdf!
- This implies there is no unique global optimum for the log likelihood function.
- Our goal is to find a good overall likelihood.

Algorithmic Issues: Optimization

- The MLE $\boldsymbol{\theta}_{\mathrm{ML}}$ is found by maximizing the log likelihood

$$\log p(\mathbf{x}_1, \ldots, \mathbf{x}_n \mid \boldsymbol{\theta}) = \sum_{i=1}^{n} \log \sum_{k=1}^{K} \boldsymbol{\pi}(k) p(\mathbf{x}_i \mid \boldsymbol{\eta}_k).$$

- log cannot change order with sum, making the above optimization problem (not concave) hard to solve. E.g., even if $\boldsymbol{\pi}$ is known, taking gradient and finding root, we need to solve:

$$\sum_{i=1}^{n} \sum_{k=1}^{K} \frac{\boldsymbol{\pi}(k)}{\sum_{k'=1}^{K} \boldsymbol{\pi}[k'] p_{k'}(\mathbf{x}_i \mid \boldsymbol{\theta})} \frac{\partial p(\mathbf{x}_i \mid \boldsymbol{\eta}_k)}{\partial \boldsymbol{\theta}} = 0.$$

- However, if we also observe the latent variables $z_1, \ldots, z_n$, the likelihood is

$$\log p((\mathbf{x}_1, z_1), \ldots, (\mathbf{x}_n, z_n) \mid \boldsymbol{\theta}) = \sum_{i=1}^{n} \left(\log \boldsymbol{\pi}[z_i] + \log p(\mathbf{x}_i \mid \boldsymbol{\eta}_{z_i})\right),$$

which is much easier to maximize!

GMM

- Observe $\mathbf{x}_1, \ldots, \mathbf{x}_n \in \mathbb{R}^D$ generated by a mixture of K Gaussian distributions $\mathcal{N}(\cdot \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, $k = 1, \ldots, K$, with

$$p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k) \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k),$$

  where $\boldsymbol{\theta} = (\boldsymbol{\pi}(k), \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)_{k=1}^K$.

- Let $\mathbf{y}_i = (\mathbf{x}_i, z_i)$ be the "complete" data, where $z_i \in \{1, \ldots, K\}$ indicates which Gaussian distribution $\mathbf{x}_i$ is drawn from:

$$\log p(\mathbf{y}_1, \ldots, \mathbf{y}_n \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \sum_{i:z_i=k} (\log \boldsymbol{\pi}(k) + \log \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)).$$

- Therefore, can find MLE of $(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$, $k = 1, \ldots, K$, in the usual way:

$$\hat{\boldsymbol{\mu}}_k = \frac{1}{n} \sum_{i:z_i=k} \mathbf{x}_i$$

$$\hat{\boldsymbol{\Sigma}}_k = \frac{1}{n} \sum_{i:z_i=k} (\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)(\mathbf{x}_i - \hat{\boldsymbol{\mu}}_k)^\mathsf{T}$$

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Basic Idea

- Suppose $p(\mathbf{y} \mid \boldsymbol{\theta})$ is easy to maximize.
- But we observe $\mathbf{x} = T(\mathbf{y})$ instead. $\mathbf{x}$ is the incomplete data and $\mathbf{y}$ is the complete data.
- The MLE is found by

$$\boldsymbol{\theta}_{\mathrm{ML}} = \arg\max_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} \log p(\mathbf{x} \mid \boldsymbol{\theta}).$$

- However, $\log p(\mathbf{x} \mid \boldsymbol{\theta})$ may be hard to compute or maximize.
- If we observe the complete data $\mathbf{y}$, we can maximize $\log p(\mathbf{y} \mid \boldsymbol{\theta})$. But now, $\mathbf{y}$ is unknown.
- Instead, let's try to maximize

$$\mathbb{E}_{p(\mathbf{y} \mid \mathbf{x}, \widehat{\boldsymbol{\theta}})}\left[\log p(\mathbf{y} \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}, \widehat{\boldsymbol{\theta}}\right],$$

where $\widehat{\boldsymbol{\theta}}$ is a "guess" of $\boldsymbol{\theta}$.

Expectation-Maximization (EM) Algorithm

1. Pick an initial guess $\boldsymbol{\theta}^{(0)}$.
2. **E step**: At iteration $m+1$, evaluate

$$
\begin{aligned}
Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) &= \mathbb{E}_{p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})}\left[\log p(\mathbf{y} \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)}\right] \\
&= \int \log p(\mathbf{y} \mid \boldsymbol{\theta}) \cdot p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \,\mathrm{d}\mathbf{y}.
\end{aligned}
$$

3. **M step**: Find $(m+1)$th guess of $\boldsymbol{\theta}$ as

$$
\boldsymbol{\theta}^{(m+1)} = \operatorname*{arg\,max}_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}).
$$

4. Repeat **E step** and **M step** until a convergence criterion.

Example: GMM

- $\mathbf{y} = (\mathbf{x}_i, z_i)_{i=1}^n$, where z_i, $i = 1, \ldots, n$, are the latent variables.
- We can interpret $\mathbf{x} = (\mathbf{x}_i)_{i=1}^n = T(\mathbf{y})$ as the incomplete data.
- Recall that $p(\mathbf{y} \mid \boldsymbol{\theta})$ is easy to maximize.
- **E step**: Evaluate

$$
\begin{aligned}
Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) &= \mathbb{E}_{p(\mathbf{y}|\mathbf{x},\boldsymbol{\theta}^{(m)})}\left[\log p(\mathbf{y} \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)}\right] \\
&= \mathbb{E}_{p(\mathbf{y}|\mathbf{x},\boldsymbol{\theta}^{(m)})}\left[\sum_{i=1}^n \log p(\mathbf{x}_i, z_i \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)}\right] \\
&= \sum_{i=1}^n \mathbb{E}_{p(z_i|\mathbf{x}_i,\boldsymbol{\theta}^{(m)})}\left[\log p(\mathbf{x}_i, z_i \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}_i, \boldsymbol{\theta}^{(m)}\right].
\end{aligned}
$$

EM for GMM (Supplementary)

- **E step**. Complete data: $\mathbf{y} = (\mathbf{x}_i, z_i)_{i=1}^n$, where $(\mathbf{x}_i, z_i)$ are i.i.d.

$$
\begin{aligned}
&Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \\
&= \sum_{i=1}^n \mathbb{E}_{\boldsymbol{\theta}^{(m)}} \left[\log p(\mathbf{x}_i, z_i \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}_i, \boldsymbol{\theta}^{(m)} \right] \\
&= \sum_{i=1}^n \sum_{k=1}^K p(z_i = k \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)}) \mathbb{E}_{\boldsymbol{\theta}^{(m)}} \left[\log p(\mathbf{x}_i, z_i \mid \boldsymbol{\theta}) \,\middle|\, z_i = k, \mathbf{x}_i, \boldsymbol{\theta}^{(m)} \right] \\
&= \sum_{i=1}^n \sum_{k=1}^K r_{ik}^{(m)} \log\{\boldsymbol{\pi}(k) \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\} \\
&:= \sum_{i=1}^n \sum_{k=1}^K r_{ik}^{(m)} \log \boldsymbol{\pi}(k) + \sum_{i=1}^n \sum_{k=1}^K r_{ik}^{(m)} \log \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)
\end{aligned}
$$

EM for GMM (Supplementary)

From Bayes' theorem,

$$\text{responsibility } r_{ik}^{(m)} = p(z_i = k \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$$
$$= \frac{p(z_i = k \mid \boldsymbol{\theta}^{(m)}) p(\mathbf{x}_i \mid z_i, \boldsymbol{\theta}^{(m)})}{p(\mathbf{x}_i \mid \boldsymbol{\theta}^{(m)})}$$
$$= \frac{\boldsymbol{\pi}^{(m)}(k) \mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_k^{(m)}, \boldsymbol{\Sigma}_k^{(m)}\right)}{\sum_{k'} \boldsymbol{\pi}^{(m)}(k') \mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_{k'}^{(m)}, \boldsymbol{\Sigma}_{k'}^{(m)}\right)}.$$

Let $n_k^{(m)} = \sum_{i=1}^n r_{ik}^{(m)}$, then

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \sum_{k=1}^K n_k^{(m)} \log \boldsymbol{\pi}(k) - \frac{1}{2} \sum_{k=1}^K n_k^{(m)} \log |\boldsymbol{\Sigma}_k|$$
$$- \frac{1}{2} \sum_{i=1}^n \sum_{k=1}^K r_{ik}^{(m)} (\mathbf{x}_i - \boldsymbol{\mu}_k)^\intercal \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k).$$

EM for GMM (Supplementary)

M step.

- $$\begin{aligned}
\max_{\boldsymbol{\theta}=(\boldsymbol{\pi},\{\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k\}_{k=1}^K)} \quad & Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \\
\text{s. t.} \quad & \sum_{k=1}^K \boldsymbol{\pi}(k) = 1,\ \boldsymbol{\pi}(k) \geq 0,\ k = 1, \ldots, K, \\
& \boldsymbol{\Sigma}_k \succ \mathbf{0},\ k = 1, \ldots, K.
\end{aligned}$$

- To optimize for $\boldsymbol{\pi}$, we first form the Lagrangian

$$\sum_{k=1}^K n_k^{(m)} \log \boldsymbol{\pi}(k) - \lambda \sum_{k=1}^K \boldsymbol{\pi}(k),$$

and taking partial derivatives w.r.t. $\boldsymbol{\pi}(k)$,

$$n_k^{(m)} \frac{1}{\boldsymbol{\pi}(k)} - \lambda = 0 \Rightarrow \boldsymbol{\pi}(k) = \frac{1}{\lambda} n_k^{(m)}.$$

With the constraint $\sum_k \boldsymbol{\pi}(k) = 1$, we have $\lambda = \sum_{k=1}^K n_k^{(m)} = n$. Therefore,

$$\boldsymbol{\pi}^{(m+1)}(k) = \frac{n_k^{(m)}}{n}.$$

EM for GMM (Supplementary)

- To find $\boldsymbol{\mu}_k^{(m+1)}$, we take

$$\frac{\partial Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})}{\partial \boldsymbol{\mu}_k} = \boldsymbol{\Sigma}_k^{-1} \sum_{i=1}^{n} r_{ik}^{(m)} (\mathbf{x}_i - \boldsymbol{\mu}_k)$$
$$= \boldsymbol{\Sigma}_k^{-1} \left(\sum_{i=1}^{n} r_{ik}^{(m)} \mathbf{x}_i - n_k^{(m)} \boldsymbol{\mu}_k \right).$$

Setting to zero and solving, we have

$$\boldsymbol{\mu}_k^{(m+1)} = \frac{1}{n_k^{(m)}} \sum_{i=1}^{n} r_{ik}^{(m)} \mathbf{x}_i.$$

EM for GMM (Supplementary)

- To find $\boldsymbol{\Sigma}_k^{(m+1)}$, we take

$$\begin{aligned}\frac{\partial Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})}{\partial \boldsymbol{\Sigma}_k} &= -\frac{1}{2} n_k^{(m)} \frac{\partial}{\partial \boldsymbol{\Sigma}_k} \log |\boldsymbol{\Sigma}_k| - \frac{1}{2} \sum_{i=1}^{n} r_{ik}^{(m)} \frac{\partial}{\partial \boldsymbol{\Sigma}_k} (\mathbf{x}_i - \boldsymbol{\mu}_k)^\intercal \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k) \\ &= -\frac{1}{2} n_k^{(m)} \boldsymbol{\Sigma}_k^{-1} + \frac{1}{2} \sum_{i=1}^{n} r_{ik}^{(m)} \boldsymbol{\Sigma}_k^{-1} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^\intercal \boldsymbol{\Sigma}_k^{-1}.\end{aligned}$$

Setting to zero and solving, we have

$$\boldsymbol{\Sigma}_k^{(m+1)} = \frac{1}{n_k^{(m)}} \sum_{i=1}^{n} r_{ik}^{(m)} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(m+1)})(\mathbf{x}_i - \boldsymbol{\mu}_k^{(m+1)})^\intercal \succ \mathbf{0}.$$

EM for GMM: Summary

Require: $\boldsymbol{\pi}^{(0)}$, $\boldsymbol{\mu}_k^{(0)}$, $\boldsymbol{\Sigma}_k^{(0)}$, $k = 1, \ldots, K$.

$$L^{(0)} = \frac{1}{n} \sum_{i=1}^{n} \log \left(\sum_{k=1}^{K} \boldsymbol{\pi}^{(0)}(k) \mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_k^{(0)}, \boldsymbol{\Sigma}_k^{(0)} \right) \right).$$

1: **repeat**

2: E step: For $i = 1, \ldots, n$, $k = 1, \ldots, K$, compute

$$\text{responsibility } r_{ik}^{(m)} = \frac{\boldsymbol{\pi}^{(m)}(k) \mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_k^{(m)}, \boldsymbol{\Sigma}_k^{(m)} \right)}{\sum_{k'} \boldsymbol{\pi}^{(m)}(k') \mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_{k'}^{(m)}, \boldsymbol{\Sigma}_{k'}^{(m)} \right)}, \tag{3.3}$$

$$n_k^{(m)} = \sum_{i=1}^{n} r_{ik}. \tag{3.4}$$

EM for GMM: Summary (cont.)

3: M step: For $k = 1, \ldots, K$, compute

$$\boldsymbol{\pi}^{(m+1)}(k) = \frac{n_k^{(m)}}{n}, \tag{3.5}$$

$$\boldsymbol{\mu}_k^{(m+1)} = \frac{1}{n_k^{(m)}} \sum_{i=1}^{n} r_{ik}^{(m)} \mathbf{x}_i, \tag{3.6}$$

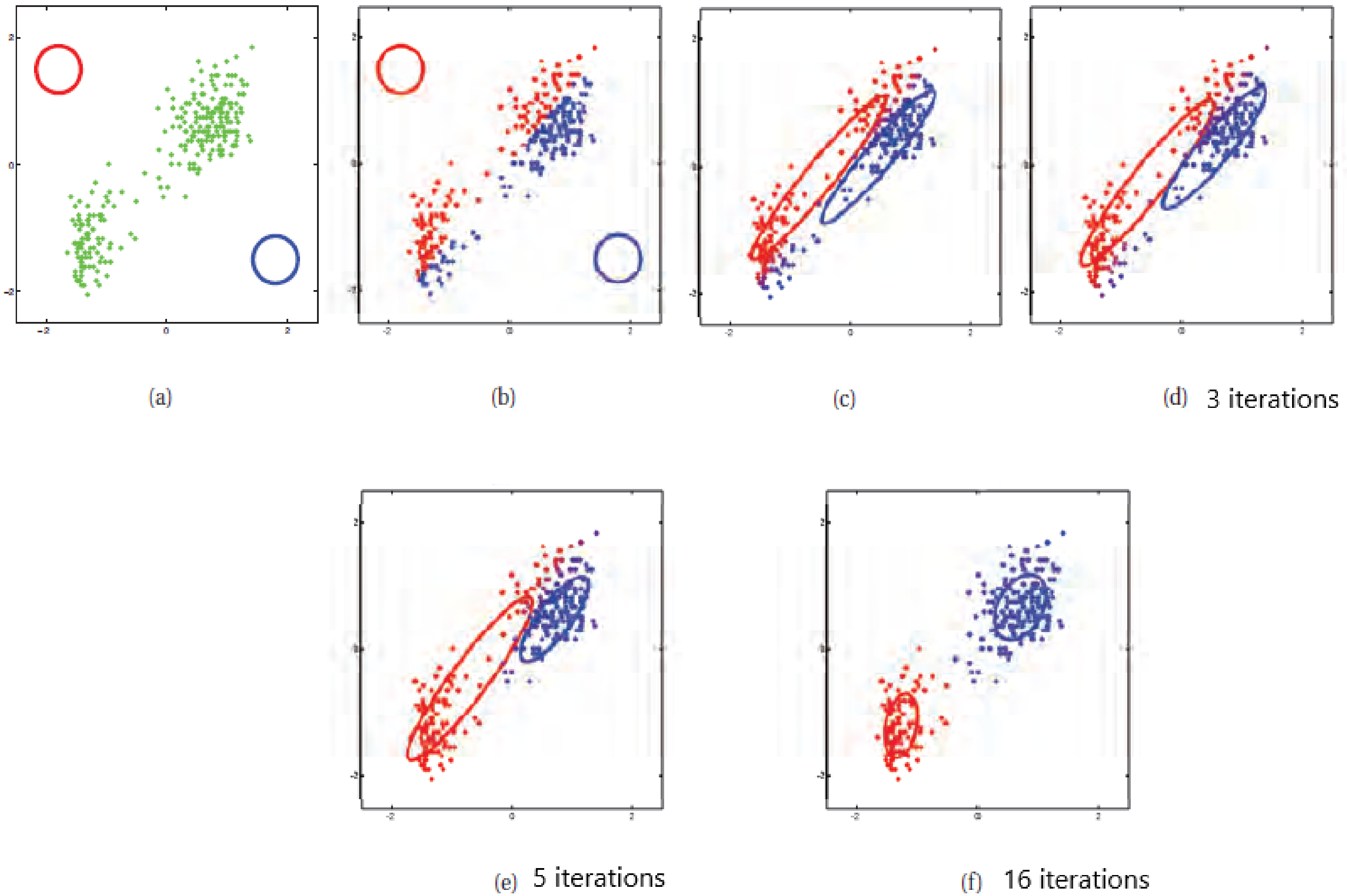
$$\boldsymbol{\Sigma}_k^{(m+1)} = \frac{1}{n_k^{(m)}} \sum_{i=1}^{n} r_{ik}^{(m)} (\mathbf{x}_i - \boldsymbol{\mu}_k^{(m+1)})(\mathbf{x}_i - \boldsymbol{\mu}_k^{(m+1)})^\intercal. \tag{3.7}$$

4: Compute

$$L^{(m+1)} = \frac{1}{n} \sum_{i=1}^{n} \log\left(\sum_{k=1}^{K} \boldsymbol{\pi}^{(m+1)}(k) \mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_k^{(m+1)}, \boldsymbol{\Sigma}_k^{(m+1)}\right)\right).$$

5: **until** $\left|L^{(m+1)} - L^{(m)}\right| \leq \epsilon$.

Example: GMM



- : 03_mix_gauss_demo_faithful.ipynb

K-Means

- GMM with $\mathbf{\Sigma}_k = \sigma^2 \mathbf{I}$ and $\boldsymbol{\pi}(k) = 1/K$ are fixed and known.
- Only $\boldsymbol{\mu}_k$, $k = 1, \ldots, K$, are to be inferred. Let $\boldsymbol{\theta} = (\boldsymbol{\mu}_1, \ldots, \boldsymbol{\mu}_K)$.
- In E step, we fix $r_{ik_i}^{(m)} = 1$ if

$$k_i = \arg\min_k \|\mathbf{x}_i - \boldsymbol{\mu}_k^{(m)}\|^2,$$

  and $r_{ik}^{(m)} = 0$ for all $k \neq k_i$ (this is called "hard EM"), i.e., for each observation $\mathbf{x}_i$, find the cluster center $\boldsymbol{\mu}_k^{(m)}$ closest to it in Euclidean distance. We then have

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \|\mathbf{x}_i - \boldsymbol{\mu}_{k_i}\|^2 + \text{const}\,.$$

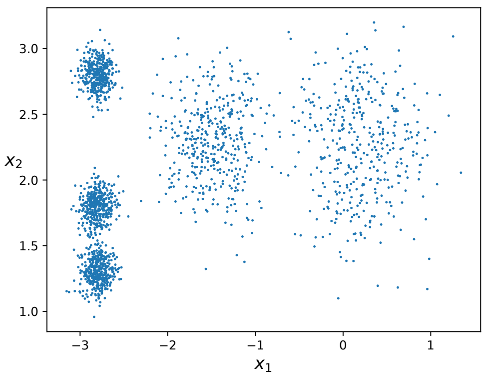
- M step: to maximize $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$, (note that this is a least squares optimization) we obtain

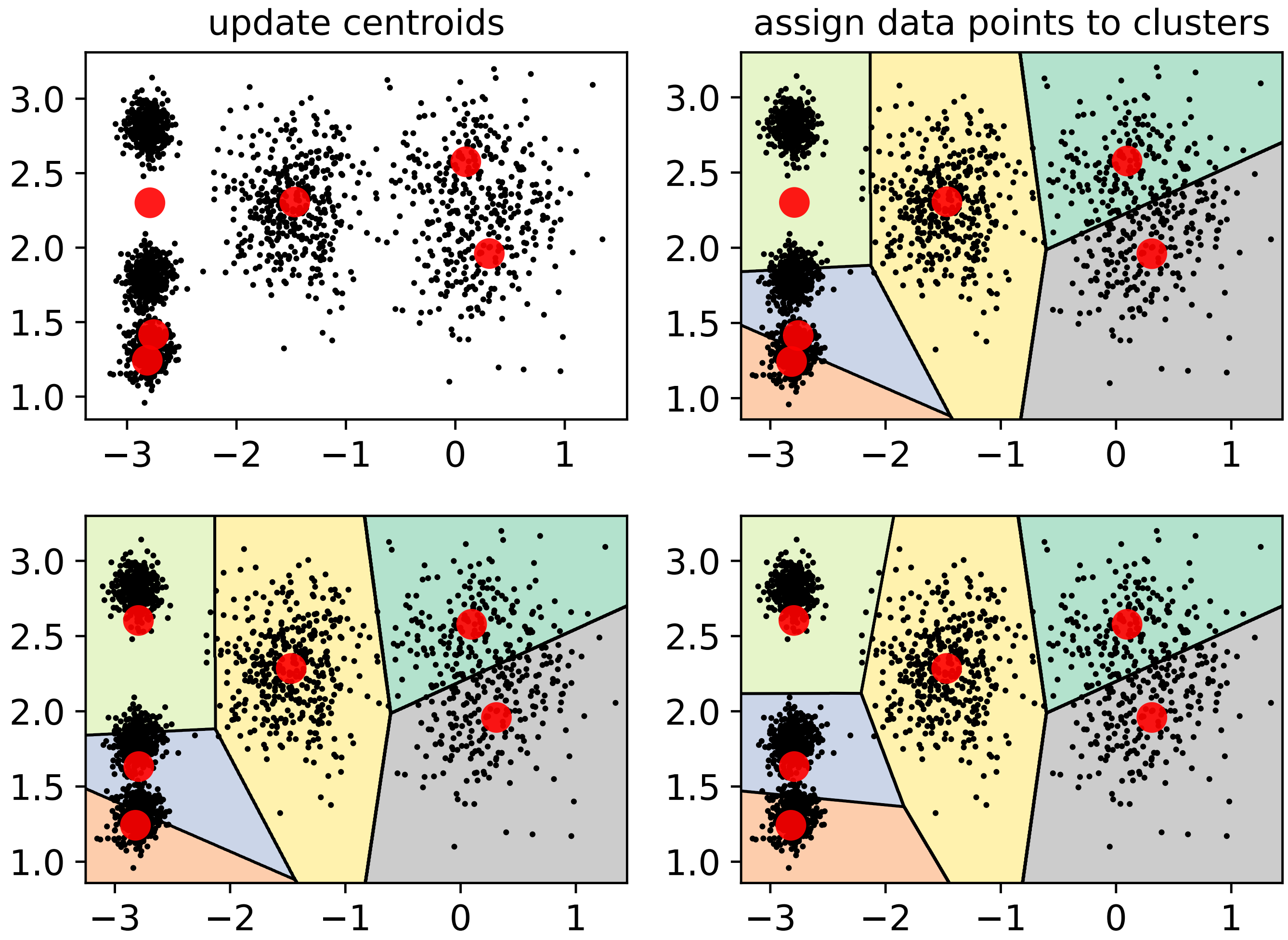
$$\boldsymbol{\mu}_k^{(m+1)} = \frac{1}{N_k} \sum_{i:k_i=k} \mathbf{x}_i, \text{ where } N_k = \sum_{i=1}^{n} \mathbf{1}\{k_i = k\}.$$

- I.e., update each cluster center with the mean of the points assigned to it in the E step.

Python: K-Means

- : 03_kmeans_voronoi.ipynb





MAP Estimation

- The EM algorithm does not resolve the singularity issue of MLE.
- MLE of a GMM can be singular if one of the $\mathbf{\Sigma}_k$ is close to zero.
- Bayesian estimation: impose a prior distribution on $\boldsymbol{\theta} \Rightarrow$ penalize values of $\mathbf{\Sigma}_k$ close to zero.
- The posterior distribution given the data $\mathbf{x}$ is

$$p(\boldsymbol{\theta} \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{p(\mathbf{x})}.$$

- MAP estimate is

$$\boldsymbol{\theta}_{MAP} = \operatorname*{arg\,max}_{\boldsymbol{\theta} \in \mathbf{\Theta}} (\log p(\mathbf{x} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})).$$

EM Algorithm for MAP

Complete data is $\mathbf{y}$, observe $\mathbf{x} = T(\mathbf{y})$.

1. Pick an initial guess $\boldsymbol{\theta}^{(0)}$.
2. **E step**: At iteration $m+1$, evaluate

$$Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \int p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \log p(\mathbf{y} \mid \boldsymbol{\theta}) \, \mathrm{d}\mathbf{y}$$

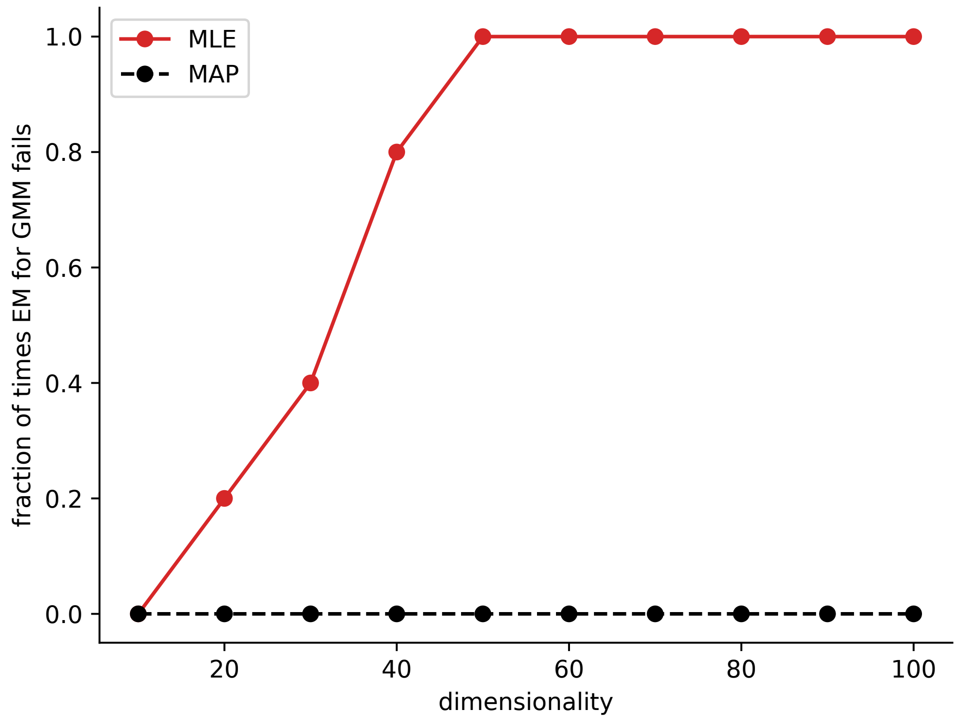
3. **M step**: Find $(m+1)$th guess of $\boldsymbol{\theta}$ as

$$\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta} \in \boldsymbol{\theta}} (Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) + \log p(\boldsymbol{\theta}))$$

4. Repeat E and M steps.

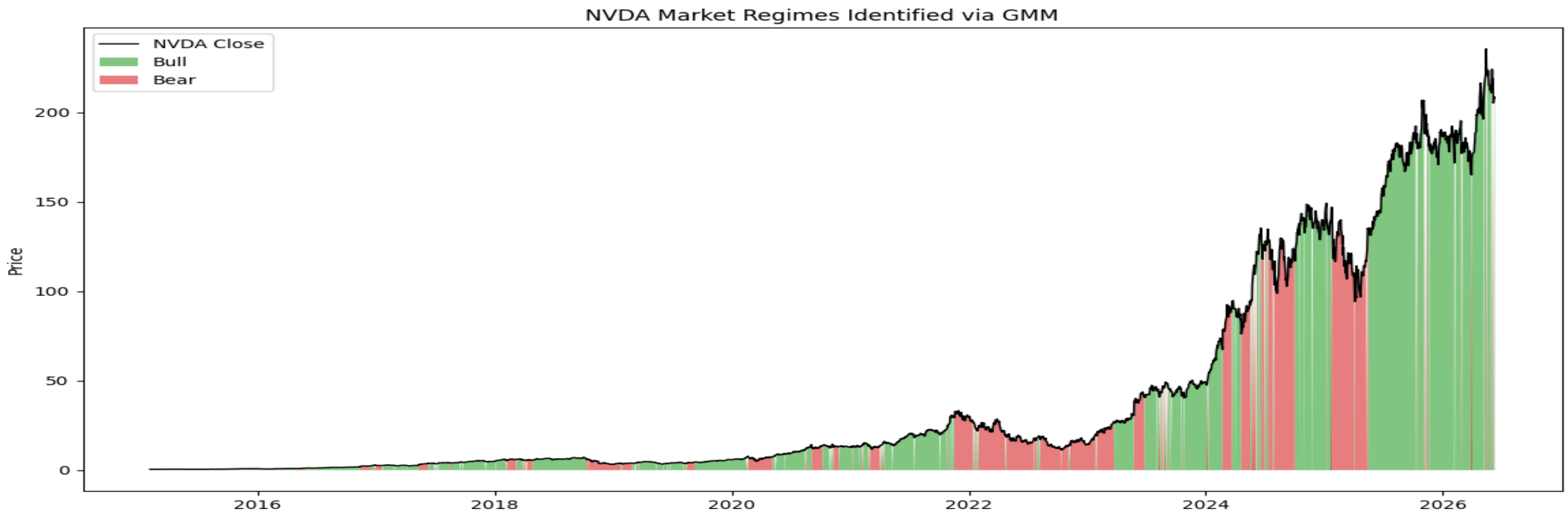
EM for MAP estimation of GMM

- Add prior distributions on $\boldsymbol{\pi}$, $\boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_k$. See [M12].
  - As dimension of $\mathbf{x}$ increases, the number of parameters in $\boldsymbol{\theta}$ increases.
  - The MLE (using the EM algorithm) may encounter numerical issues due to singular matrices.
  - Using the EM with MAP estimation can help to regularize the parameters.



A Real Application

- Stock price data from Yahoo Finance: `https://finance.yahoo.com/`
- Believe that the stock price data is generated by a mixture of two Gaussian distributions: one for the "bull market" and one for the "bear market".
- Use the EM algorithm to estimate the parameters of the two Gaussian distributions and the mixing coefficients.
- 03_market_gmm.ipynb



Summary

- Mixture model: $p(\mathbf{x} \mid \boldsymbol{\theta}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k) p_k(\mathbf{x} \mid \boldsymbol{\theta})$, GMM
- EM algorithm for finding the MLE.
  - E step: compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathbb{E}_{\mathbf{y} \sim p(\cdot \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \left[\log p(\mathbf{y} \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)} \right]$
  - M step: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta} \in \boldsymbol{\Theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$
- K-means as a special case of EM on GMM.
- EM algorithm for finding the MAP estimate.

Outline

1. Mixture Models
2. EM Algorithm
3. Some Practice Problems
4. Background Reading: Theoretical Basis (Supplementary)

Some Practice Problems

1. Consider the GMM. Show that

$$\mathbb{E}[\mathbf{x}] = \sum_{k=1}^{K} \boldsymbol{\pi}(k)\boldsymbol{\mu}_k,$$

$$\mathrm{cov}(\mathbf{x}) = \sum_{k=1}^{K} \boldsymbol{\pi}(k)(\boldsymbol{\Sigma}_k + \boldsymbol{\mu}_k\boldsymbol{\mu}_k^{\intercal}) - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^{\intercal}.$$

Solution: Let z be the latent variable, then

$$\mathbb{E}[\mathbf{x}] = \mathbb{E}[\mathbb{E}[\mathbf{x} \mid z]] = \sum_{k=1}^{K} \boldsymbol{\pi}(k)\mathbb{E}[\mathbf{x} \mid z = k] = \sum_{k=1}^{K} \boldsymbol{\pi}(k)\boldsymbol{\mu}_k.$$

$$\begin{aligned}\mathrm{cov}(\mathbf{x}) &= \mathbb{E}[(\mathbf{x} - \mathbb{E}\mathbf{x})(\mathbf{x} - \mathbb{E}\mathbf{x})^{\intercal}] = \mathbb{E}[\mathbf{x}\mathbf{x}^{\intercal}] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^{\intercal} \\ &= \sum_{k=1}^{K} \boldsymbol{\pi}(k)\mathbb{E}[\mathbf{x}\mathbf{x}^{\intercal} \mid z = k] - \mathbb{E}[\mathbf{x}]\mathbb{E}[\mathbf{x}]^{\intercal}. \qquad (3.8)\end{aligned}$$

Some Practice Problems (cont.)

For each $z = k$,

$$
\begin{aligned}
\boldsymbol{\Sigma}_k &= \mathbb{E}[(\mathbf{x} - \boldsymbol{\mu}_k)(\mathbf{x} - \boldsymbol{\mu}_k)^\intercal \mid z = k] \\
&= \mathbb{E}[\mathbf{x}\mathbf{x}^\intercal \mid z = k] - \mathbb{E}[\boldsymbol{\mu}_k \mathbf{x}^\intercal \mid z = k] - \mathbb{E}[\mathbf{x}\boldsymbol{\mu}_k^\intercal \mid z = k] + \boldsymbol{\mu}_k\boldsymbol{\mu}_k^\intercal \\
&= \mathbb{E}[\mathbf{x}\mathbf{x}^\intercal \mid z = k] - \boldsymbol{\mu}_k\boldsymbol{\mu}_k^\intercal - \boldsymbol{\mu}_k\boldsymbol{\mu}_k^\intercal + \boldsymbol{\mu}_k\boldsymbol{\mu}_k^\intercal \\
&= \mathbb{E}[\mathbf{x}\mathbf{x}^\intercal \mid z = k] - \boldsymbol{\mu}_k\boldsymbol{\mu}_k^\intercal.
\end{aligned}
$$

Therefore, $\mathbb{E}[\mathbf{x}\mathbf{x}^\intercal \mid z = k] = \boldsymbol{\Sigma}_k + \boldsymbol{\mu}_k\boldsymbol{\mu}_k^\intercal$. Substituting into (3.8), we obtain the desired result.

Some Practice Problems (cont.)

2. The following code snippet is related to an EM algorithm implementation of GMM. Explain what each line is computing, using equations, starting from Line 13.

```
1  # rawdata is a text file with 500 pairs of real values arranged in rows
2  X = np.loadtxt(rawdata)
3  X = (X - X.mean(axis=0)) / (X.std(axis=0))
4  mu1 = np.array([-1.5, 1.5])
5  mu2 = np.array([1.5, -1.5])
6  Sigma1 = np.identity(2) * 0.1
7  Sigma2 = np.identity(2) * 0.1
8
9  pi = [0.5, 0.5]
10 mu = [mu1, mu2]
11 Sigma = [Sigma1, Sigma2]
12
13 Ns = [multivariate_normal(mean=mu_i, cov=Sigma_i) for mu_i, Sigma_i in zip(mu, Sigma)]
14 elements = [pi_i * Ni.pdf(X) for pi_i, Ni in zip(pi, Ns)]
15 elements = elements / np.sum(elements, axis=0)
```

Solution:

Line 13: For each mixture component, creates a normal distribution object: $\mathcal{N}\left(\cdot \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i\right)$

Some Practice Problems (cont.)

Line 14: For each row i in X, the data is $\mathbf{x}_i$. Computes the probabilities for $k = 1, 2$:

$$\boldsymbol{\pi}^{(m)}(k)\mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_k^{(m)}, \boldsymbol{\Sigma}_k^{(m)}\right)$$

Line 15: Computes the responsibility

$$r_{ik}^{(m)} = \frac{\boldsymbol{\pi}^{(m)}(k)\mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_k^{(m)}, \boldsymbol{\Sigma}_k^{(m)}\right)}{\sum_{k'} \boldsymbol{\pi}^{(m)}(k')\mathcal{N}\left(\mathbf{x}_i \,\middle|\, \boldsymbol{\mu}_{k'}^{(m)}, \boldsymbol{\Sigma}_{k'}^{(m)}\right)}.$$

Some Practice Problems (cont.)

3. On a walking-cum-cycling path, the pedestrian-to-cyclist ratio is $4:1$. A person is considered *vulnerable* if they are below 10, above 65, or handicapped. The probability of a pedestrian being vulnerable is θ_0, and for a cyclist, it is θ_1. Hua surveys 100 individuals, recording whether each is vulnerable $(x_i = 1)$ or not $(x_i = 0)$ for $i = 1, \ldots, 100$, but does not record if they are pedestrians or cyclists.

   (i) Write down the log-likelihood function for the data collected by Hua by filling in the blank boxes in the expression below.

   $$\log p(x_1, \ldots, x_{100} \mid \theta_0, \theta_1) = \sum_{i=1}^{100} \log\left(0.8 \cdot \square^{x_i} \cdot (1 - \square)^{\square} + 0.2 \cdot \square^{\square} \cdot (1 - \square)^{\square}\right)$$

   (ii) What is a suitable algorithm that Hua should use to find the MLEs of θ_0 and θ_1? Justify your answer.

Some Practice Problems (cont.)

Solution:

(i)

$$\log p(x_1, \dots, x_{100} \mid \theta_0, \theta_1) = \sum_{i=1}^{100} \log\left(0.8 \cdot \theta_0^{x_i}(1-\theta_0)^{1-x_i} + 0.2 \cdot \theta_1^{x_i}(1-\theta_1)^{1-x_i}\right)$$

(ii) This is a mixture model and the summation and $\log$ are not interchangeable, which makes directly taking derivatives and solving for the root difficult. Therefore, Hua should use the EM algorithm to find the MLEs of θ_0 and θ_1.

Outline

1. Mixture Models
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Monotonicity

- Is EM guaranteed to find the MLE? No.

- Can get stuck in local maximum. In practice, start EM from multiple random initial guesses, and choose the result with the highest likelihood.

- However, at each iteration, the log likelihood $l(\boldsymbol{\theta}) = \log p(\mathbf{x} \mid \boldsymbol{\theta})$ is non-decreasing:

$$l(\boldsymbol{\theta}^{(m+1)}) \geq l(\boldsymbol{\theta}^{(m)})$$

  $\Rightarrow$ if $l(\boldsymbol{\theta})$ is bounded above, EM algorithm produces a sequence of estimates that converges to a (possibly local) maximum.

Monotonicity (cont.)

Proof: Recall $\mathbf{x} = T(\mathbf{y})$ is the observed incomplete data and $\mathbf{y}$ is the complete data.

$$
\begin{aligned}
l(\boldsymbol{\theta}) &= \log p(\mathbf{x} \mid \boldsymbol{\theta}) \\
&= \log \int p(\mathbf{x}, \mathbf{y} \mid \boldsymbol{\theta}) \, \mathrm{d}\mathbf{y} \\
&= \log \int_{T(\mathbf{y})=\mathbf{x}} p(\mathbf{y} \mid \boldsymbol{\theta}) \, \mathrm{d}\mathbf{y} \\
&= \log \int_{T(\mathbf{y})=\mathbf{x}} \frac{p(\mathbf{y} \mid \boldsymbol{\theta})}{p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \, \mathrm{d}\mathbf{y} \\
&\overset{\text{Jensen}}{\geq} \mathbb{E}_{\mathbf{y} \sim p(\cdot \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \left[\log \frac{p(\mathbf{y} \mid \boldsymbol{\theta})}{p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)} \right] \\
&= Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - \mathbb{E}_{\mathbf{y} \sim p(\cdot \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \left[\log p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)} \right].
\end{aligned}
$$

Recall $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathbb{E}_{\mathbf{y} \sim p(\cdot \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \left[\log p(\mathbf{y} \mid \boldsymbol{\theta}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)} \right]$.

Monotonicity (cont.)

By the M step, we have $Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) \geq Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$, so

$$
\begin{aligned}
l(\boldsymbol{\theta}^{(m+1)}) &\geq Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)}) - \mathbb{E}_{\boldsymbol{\theta}^{(m)}}\left[\log p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \,\middle|\, \mathbf{x}, \boldsymbol{\theta}^{(m)}\right] \\
&= \int_{T(\mathbf{y})=\mathbf{x}} p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \log \frac{p(\mathbf{y} \mid \boldsymbol{\theta}^{(m)})}{p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \,\mathrm{d}\mathbf{y} \\
&= \int p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \log \frac{p(\mathbf{x}, \mathbf{y} \mid \boldsymbol{\theta}^{(m)})}{p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)})} \,\mathrm{d}\mathbf{y} \\
&= \int p(\mathbf{y} \mid \mathbf{x}, \boldsymbol{\theta}^{(m)}) \log p(\mathbf{x} \mid \boldsymbol{\theta}^{(m)}) \,\mathrm{d}\mathbf{y} \\
&= \log p(\mathbf{x} \mid \boldsymbol{\theta}^{(m)}) \\
&= l(\boldsymbol{\theta}^{(m)})
\end{aligned}
$$